GCD
HCF ( Greatest common divisor )

$x = \gcd(a, b)$ = Greatest no which divides a & b

$a \% x == 0$ && $b \% x == 0$ , $x$ is highest number

$\gcd(15, 25) = 5$

| 15 | 25 |
|---|---|
| 1 | 1 |
| 3 | 5 |
| 5 | 25 |
| 15 | |

$\gcd(10, -25) = 5$

| 10 | -25 |
|---|---|
| 1 | 1 |
| 2 | 5 |
| 5 | 25 |
| 10 | |

$\gcd(0, 8) = 8$

| 0 | 8 |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 4 |
| ⋮ | 8 |
| 8 | |
| ⋮ | |
| 100 | |

$\gcd(0, -10) = 10$

| 0 | -10 |
|---|---|
| ⋮ | 1 |
| | 2 |
| | 5 |
| 10 | 10 |

## Properties of gcd

1. $\gcd(a, b) = \gcd(b, a)$
2. $\gcd(a, b) = \gcd(|a|, |b|)$
3. $\gcd(0, b) = b$

   $\hookrightarrow$ if $(b == 0)$ $\gcd(0,0) =$ NA

4. $\gcd(a, b, c) = \gcd(a, \gcd(b, c))$

   Associative

   $= \gcd(b, \gcd(a, c))$

   $= \gcd(c, \gcd(a, b))$

$\gcd(8, 9) = 1$

| 8 | 9 |
|---|---|
| 1 | 1 |
| 2 | 3 |
| 4 | 9 |
| 8 | |

Q. Given A, B. Find gcd(A, B). Both are not zero

A, B > 0

$gcd \in [1, \min(A,B)]$

```
for( i = min(A,B) ; i ≥ 1 ; i-- )
{
    if( B%i == 0 && A%i == 0)
    {   return i
}
```

TC : $O(\min(A, B))$

SC : $O(1)$

10          25

↓

$\sqrt{N}$

1

2

5

10

Idea: Calculate factors of smaller numbers.

gcd = Find max factor, which divides the larger number.

TC: $O(\sqrt{min(A,B)})$

SC: $O(\sqrt{min(A,B)})$

$\downarrow$

$O(1)$

Calculate factors of smaller numbers

$1$ $\quad\quad$ $\sqrt{N}$

$i$ $\quad\quad$ $N/i$

$O(\sqrt{N})$

If $a, b$ $\quad\quad$ $a > b$

$gcd(a, b) = x$ $\quad\quad$ $a \% x == 0$ && $b \% x == 0$

$\Downarrow$

$gcd(a-b, b) = x$

$= (a-b) \% x$

$(a \% x - b \% x + x) \% x$

$= 0$

$a > b$ A

$\gcd(a, b) = \gcd(a-b, b)$ $\quad A > b$

$= \gcd(a - b - b, b)$

$= \gcd(a-2b, b)$ $\quad a-2b > b$

$= \gcd(a-3b, b)$ $\quad a-3b > b$

$= \gcd(a-4b, b)$

$x$ : max multiple of $b$. ST $a - xb \geq 0$

$= \gcd(a - xb, b)$

$= \gcd(a \% b, b)$

$10 \% 4 = 10 - (\text{max multiple of } 4 \leq 10)$

$= 10 - 8 = 2$

$a > b$

$\gcd(a, b) = \gcd(a \% b, b)$

$gcd(21, 6) = gcd(21 \% 6, 6)$

$= gcd(3, 6)$

3

$= gcd(6, 3)$

$= gcd(6 \% 3, 3)$

$= gcd(0, 3)$

$= 3$

$gcd(10, 25) = gcd(25, 10)$

$= gcd(25 \% 10, 10)$

$= gcd(5, 10)$

$= gcd(10, 5)$

$= gcd(10 \% 5, 5)$

$= gcd(0, 5)$

$= 5$

```
int gcd (int a, int b)
{
    if ( a < b )   swap(a,b)
    if ( b == 0 )  return A
    return  gcd( a % b, b )
}
```

5

gcd(10, 25)

a = 25, b = 10

gcd(25 % 10, 10)

5

gcd(5, 10)

a = 10, b = 5

gcd(10 % 5, 5)

gcd(0, 5)

a = 5  b = 0

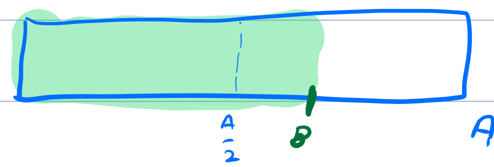
if (b == 0) return 5

gcd(a, b)  gcd(a % b, b)

$b \le \frac{A}{2}$ $\qquad\qquad$ $b > \frac{A}{2}$

$a \% b < b$

$a \% b < \frac{A}{2}$




$a \% b = a - x b$

$a \% b = a - b$

$a \% b < \frac{A}{2}$

TC: $\log_2 A + \log_2 B$

TC: $O(\log_2 \max(A, B))$

SC: $O(\log_2 \max(A, B))$

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$1 \le arr[i] \le m$

Q. Given array of size N. Find out gcd of the factorial of given number. ans % ($1e^9+7$)

Ex: [4, 3, 8, 6]

$4! = 1 \times 2 \times 3 \times 4$

$3! = 1 \times 2 \times 3$

$8! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$

$3! = 6$

$6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6$

ans = (min number)! % m

TC: O(N + m)

**Q** Given an array. Check if there exists a subsequence with gcd = 1

Ex 4, 6, 3, 8 — {4, 3} {3, 8} {4, 3, 8} — True

Ex 16, 10, 6, 15, 27 — {10, 27} {16, 15} {16, 27} — True

Ex 6, 12, 3, 18 — False

{6 15 10} $\quad$ {6, 15, 10} = 1

6, 8, 16, 32 $\quad$ False

A B C D E F G H

gcd (B, C, E, F) = 1

gcd (A, B, C, D, E, F, G, H) = 1

If gcd of array == 1
return True
else
return False

```
gcdtmp = arr[0]
for (i=1; i<N; i++)
{   gcdtmp = gcd(gcdtmp, arr[i])
```

gcd tmp

TC: O(N log(M))

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Q. Given N positive numbers.

Choose i, j (i != j)

and now if abs(arr[i] - arr[j]) is not present, Insert it in the array.

Keep on repeating this step, until no more addition can be done

Q. Find the min element in the final array?

Ex = [6, 2, 12, 8, 4, 10] ans = 2

Ex = [7, 9, 5, 2, 3, 4, 1, 6, 8] ans = 1

Ex = [14, 10, 4, 6, 2, 12, 8] ans = 2

92 − 10 = 82

82 − 10 = 72

72 − 10 = 62

[ 92, 10, 82, 72, 62 ]

52, 42, 32, 22, 12, 2, 8, 6, 4, , gcd 92,10 ]

$a - b$

$a - 2b$

$a - 3b$

$\vdots$

$a -$

12 − 10 = 2

[42, 10] = [ 42, 10, 32, 22, 12, 2 ]

8, 6, 4

[ a, b, c ]

gcd gcd

gcd

ans = gcd of array

For any 2 number, gcd of pair is going to be added eventually.

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$gcd(a, b, c, d)$

$gcd(gcd(gcd(a, b), c), d)$